MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Question # 4 of 10 (Start time: 04:14:36 PM, 05 January 2026)

Copositional events in Special Relativity have a spacelike interval.

Select the correct option

○	false
◉	true

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Question # 3 of 10 (Start time: 04:14:23 PM, 05 January 2026)

Under which operation the set of Lorentz matrices form a group called the Lorentz group?

Select the correct option

○	All of these
●	Multiplication of Matrices
○	Addition of Matrices
○	Subtraction of Matrices

MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Question # 2 of 10 (Start time: 04:14:07 PM, 05 January 2026)

Which of the following is the correct mathematical form of the second skew symmetry of Riemann tensor?

Select the correct option

◉	$R_{ijkl} + R_{iklj} + R_{iljk} = 0$
○	$R_{ijkl} = R_{klij}$
○	$R^{i}_{jkl} = -R^{i}_{jkl}$
○	$R_{ijkl} = -R_{ijlk}$

MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Question # 1 of 10 (Start time: 04:13:54 PM, 05 January 2026)

Which of the following best describes the event space in Special Relativity?

Select the correct option

○	A three-dimensional space defined by an inertial observer.
◉	A four-dimensional space combining three spatial dimensions and one time dimension.
○	A space where only spatial coordinates are considered.
○	A plane where events occur at the same time.

Question # 10 of 10 (Start time: 04:13:00 PM, 05 January 2026)

What is the expression for the relativistic mass $\hat{m}$ if m is rest mass of the particle?

Select the correct option

- $\hat{m} = \frac{m}{\sqrt{1 - \frac{v^2}{c^2}}}$
- $m = \frac{\hat{m}}{\sqrt{1 - \frac{v^2}{c^2}}}$
- $\hat{m} = \frac{mv}{\sqrt{1 - \frac{v^2}{c^2}}}$
- $\hat{m} = \frac{mc}{\sqrt{1 - \frac{v^2}{c^2}}}$

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Question # 9 of 10 (Start time: 04:12:46 PM, 05 January 2026)

What is true for unit sphere?

Select the correct option

◉	r = 1
○	r = 0
○	r = 2
○	r = -1

MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Question # 8 of 10 (Start time: 04:12:34 PM, 05 January 2026)

Simultaneous events in Special Relativity have a spacelike interval.

Select the correct option

○	false
◉	true

MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Question # 7 of 10 (Start time: 04:12:21 PM, 05 January 2026)

Which of the following is NOT an example of an inertial frame of reference?

Select the correct option

◉	An accelerating train.
○	A car moving at constant velocity.
○	A spaceship in deep space far from any gravitational influence.
○	A stationary observer on Earth ignoring Earth's rotation.

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Question # 6 of 10 (Start time: 04:12:10 PM, 05 January 2026)

In a curvilinear coordinate system, the straight lines are called _____.

Select the correct option

◉	geodesics
○	normals
○	none of these options
○	tangents

MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Question # 5 of 10 (Start time: 04:11:56 PM, 05 January 2026)

How is the Ricci scalar computed?

Select the correct option

○	By integrating the Ricci tensor over the manifold
●	By contracting the Ricci tensor with the metric tensor
○	By differentiating the Riemann tensor
○	By multiplying the metric tensor by the Riemann tensor

MTH623 - Tensor Analysis and its Applications (Quiz # 4) Quiz Start Time: 04:10 PM, 05 January 2026

Question # 4 of 10 (Start time: 04:11:43 PM, 05 January 2026) Total Marks: 1

Which of the following quantities are obtained, when τ - derivatives is replaced by t - derivatives in the expressions of the velocity and acceleration?

Select the correct option

Reload Math Equations

- Proper time and speed of light.
- Relativistic velocity and acceleration.
- Proper velocity and acceleration.
- Proper time and velocity.

MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Question # 3 of 10 (Start time: 04:11:28 PM, 05 January 2026)

The matrix form of Minkowski spacetime is a _____.

Select the correct option

○	identity matrix
◉	diagonal matrix
○	scalar matrix
○	null matrix

MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Question # 2 of 10 (Start time: 04:11:04 PM, 05 January 2026)

The curvature of the curve $C : x^i = x^i(t)$, is defined as _____.

Select the correct option

○	the norm of first order derivative of $x^i(s)$
○	the square root of first order derivative of $x^i(s)$
◉	the norm of second order derivative of $x^i(s)$
○	the square root of second order derivative of $x^i(s)$

MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Question # 1 of 10 (Start time: 04:10:49 PM, 05 January 2026)

What happens to the length of an object when it moves close to the speed of light?

Select the correct option

◉	It gets shorter.
○	It gets longer.
○	It disappears.
○	It stays the same.

MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Question # 10 of 10 (Start time: 04:17:17 PM, 05 January 2026)

Which of the following is true about an inertial frame of reference?

Select the correct option

○	It experiences non-uniform acceleration.
◉	It moves with uniform velocity or remains at rest.
○	It is stationary in space.
○	It is always at rest.

MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Question # 9 of 10 (Start time: 04:16:59 PM, 05 January 2026)

If a flash of light occurs at (0, 0, 0) and time t = 0 which of the following equation describes the expanding spherical wave front?

Select the correct option

○	$x^2+y^2+z^2= c^2$
●	$x^2+y^2+z^2= c^2t^2$
○	$x^2+y^2+z^2= c^2t$
○	$x^2+y^2+z^2= t^2$

MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Question # 8 of 10 (Start time: 04:16:06 PM, 05 January 2026)

The absolute derivative of T^i along the curve $C : x^i = x^i(t)$ is defined as:

Select the correct option

○	$\frac{\delta T^i}{\delta t} = \frac{\partial T^i}{\partial t} + \Gamma^i_{rs} T^s$
○	$\frac{\partial T^i}{\partial t} = \frac{dT^i}{dt} + \Gamma^i_{rs} T^s \frac{dx^r}{dt}$
●	$\frac{\delta T^i}{\delta t} = \frac{dT^i}{dt} + \Gamma^i_{rs} T^s \frac{dx^r}{dt}$
○	$\frac{\delta T^i}{\delta t} = \frac{\partial T^i}{\partial t} + \Gamma^i_{rs} T^s \frac{dx^r}{dt}$

MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Question # 7 of 10 (Start time: 04:15:53 PM, 05 January 2026)

Which term is often chosen to reparameterize the proper time ?

Select the correct option

○	$\tau = \frac{c}{s}$
○	$\tau = \frac{s}{c^2}$
●	$\tau = \frac{s}{c}$
○	$\tau = s \times c$

MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Question # 6 of 10 (Start time: 04:15:41 PM, 05 January 2026)

What is a four vector in the special relativity?

Select the correct option

○	A vector with four spatial components.
○	A vector in three-dimensional space.
○	A vector that only exists in non-relativistic physics.
◉	A vector with three spatial components and one time component.

MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Question # 5 of 10 (Start time: 04:15:06 PM, 05 January 2026)

The arc length parameter is defined as _____.

Select the correct option

○	$s(t) = \int_a^t \sqrt{\frac{dx^i}{dt}\frac{dx^j}{dt}}$
◉	$s(t) = \int_a^t \sqrt{\delta_{ij}\frac{dx^i}{dt}\frac{dx^j}{dt}}$
○	$s(t) = \int_a^t \sqrt{\delta_{ii}\frac{dx^i}{dt}\frac{dx^i}{dt}}$
○	$s(t) = \sqrt{\delta_{ij}\frac{dx^i}{dt}\frac{dx^j}{dt}}$